

Sums of (dependent) random variables

due to the linearity property of the expectation, we have

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

the variance of the sum of n variables is the sum of their covariances:

$$\begin{aligned}\mathrm{Var}\left(\sum_{i=1}^n X_i\right) &= \mathbb{E}\left[\left(\sum_i^n X_i - \mathbb{E}\left[\sum_i^n X_i\right]\right)^2\right] \\ &= \sum_{i=1}^n \sum_{j=1}^n \mathrm{Cov}(X_i, X_j) \\ &= \sum_{i=1}^n \mathrm{Var}(X_i) + 2 \sum_{i \neq j} \mathrm{Cov}(X_i, X_j).\end{aligned}$$

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- suppose that $X = k$, where k is some integer
- then $Z = z$ if and only if $Y = z - k$
- so the event $Z = z$ is the **intersection** of the two events

$$(X = k) \text{ and } (Y = z - k),$$

where k runs over the integers

Theorem 1 (PDF of $X + Y$)

if $f(x, y)$ denotes the PDF of (X, Y) and $Z = X + Y$ then:

$$f_Z(z) = \int_{-\infty}^{\infty} f(x, z - x) dx$$

proof: substituting $y = u - x$ we find:

$$\begin{aligned} F_Z(z) &= \int \int_{x+y \leq z} f(x, y) dx dy = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{z-x} f(x, y) dy \right] dx \\ &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^z f(x, u - x) du \right] dx = \int_{-\infty}^z \left[\int_{-\infty}^{\infty} f(x, u - x) dx \right] du \end{aligned}$$

taking the derivative of F_Z we come to the mentioned result

□

Example: Sums of Gaussian Random Variables

if X is distributed as $\mathcal{N}(\mu_X, \sigma_X^2)$, Y is distributed as $\mathcal{N}(\mu_Y, \sigma_Y^2)$, then

$$X + Y \sim \mathcal{N}(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2 + 2\sigma_{X,Y})$$

$$aX + bY \sim \mathcal{N}(a\mu_X + b\mu_Y, a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_{X,Y})$$

where $\sigma_{X,Y}$ is the covariance of X and Y

proof: see slides 25-26, 28



Sums of Independent (Continuous) Random Variables

the density of a sum of **independent** random variables is given by a convolution of the densities

Theorem 2

let X and Y be two independent random variables with density functions $f_X(x)$ and $f_Y(y)$ defined for all x . Then the sum $Z = X + Y$ is a random variable with density function $f_Z(z)$, where f_Z is the convolution of f_X and f_Y :

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx$$

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$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx$$

proof: this can be shown in two steps:

i) compute the CDF $F_Z(z) = \mathbb{P}(Z \leq z)$

ii) take the derivative $f_Z(z) = \frac{d}{dz} F_Z(z)$

Sums of Independent (Continuous) Random Variables

step i:

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{XY}(x, y) dy dx \quad (1)$$

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step ii:

$$\begin{aligned} f_Z(z) &= \frac{d}{dz} F_Z(z) = \frac{d}{dz} \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{XY}(x, y) dy dx \\ &= \int_{-\infty}^{\infty} f_{XY}(x, z-x) dx \end{aligned} \quad (2)$$

where the second equality comes from Leibniz integral rule

Sums of Independent (Continuous) Random Variables

note that if X and Y are independent random variables, then

$$f_{XY}(x, y) = f_X(x)f_Y(y),$$

therefore

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(z-x)dx$$

we conclude that the PDF of the sum of two independent random variables is the **convolution** of their individual PDFs:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(x, z-x)dx = \int_{-\infty}^{\infty} f_X(x)f_Y(z-x)dx$$



Example: PDF of Sum of Two Independent Exponential Random Variables – definitions

let $X \sim \exp(\lambda_1)$, $Y \sim \exp(\lambda_2)$ and $\lambda_1 \neq \lambda_2$:

$$f_X(x) = \begin{cases} \lambda_1 e^{-\lambda_1 x}, & \text{if } x \geq 0 \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

and

$$f_Y(y) = \begin{cases} \lambda_2 e^{-\lambda_2 y}, & \text{if } y \geq 0 \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

Example: PDF of Sum of Two Independent Exponential Random Variables – convolution

if $z > 0$, then the sum $Z = X + Y$ has pdf given by the convolution

$$\begin{aligned}f_Z(z) &= \int_{-\infty}^{\infty} f_Y(z-x)f_X(x)dx \\&= \lambda_1\lambda_2 \int_0^z e^{-\lambda_2(z-x)}e^{-\lambda_1x}dx \\&= \lambda_1\lambda_2 e^{-\lambda_2z} \int_0^z e^{-(\lambda_1-\lambda_2)x}dx \\&= \frac{\lambda_1\lambda_2}{\lambda_1-\lambda_2} e^{-\lambda_2z} (1 - e^{-(\lambda_1-\lambda_2)z}) \\&= \frac{\lambda_1\lambda_2}{\lambda_1-\lambda_2} (e^{-\lambda_2z} - e^{-\lambda_1z}),\end{aligned}\tag{5}$$

while if $z < 0$, $f_Z(z) = 0$

Sub-Gaussian random variables

Definition 1 (Sub-Gaussian random variable)

A random variable X is **sub-Gaussian** if there exist constants $C, v > 0$ such that for all $t > 0$:

$$\mathbb{P}(|X| > t) \leq Ce^{-vt^2}.$$

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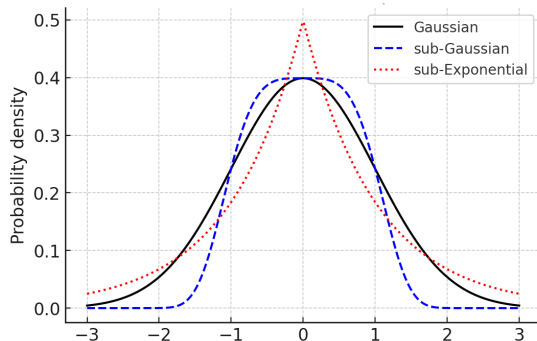
- ▶ Examples: Gaussian $\mathcal{N}(0, 1)$, Uniform on $[-1, 1]$.
- ▶ They are “light-tailed” variables – bounded or well-concentrated.

From sub-Gaussian to sub-Exponential

Tail decay

- ▶ Sub-Gaussian: $\mathbb{P}(|X| > t) \lesssim e^{-t^2}$
- ▶ Sub-Exponential: $\mathbb{P}(|X| > t) \lesssim e^{-t}$

From sub-Gaussian to sub-Exponential



Sub-Gaussian $\Rightarrow$ light tails
Sub-Exponential $\Rightarrow$ heavier tails

Random Vectors – definition

Definition 2 (random vector)

a **random vector**

$$\mathbf{x} = (X_1, \dots, X_d) \in \mathbb{R}^d$$

is a multivariate random variable, i.e. a set/collection of d random variables

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What we will focus on

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Example

Images, for instance, are represented by random vectors with millions of components (pixels).

Expectation of a Random Vector

the notion of expectation (or mean) generalizes straightforwardly to random vectors

Definition

for $\mathbf{x} = (X_1, \dots, X_d) \in \mathbb{R}^d$,

$$\mathbb{E}[\mathbf{x}] = \begin{pmatrix} \mathbb{E}[X_1] \\ \vdots \\ \mathbb{E}[X_d] \end{pmatrix} = \boldsymbol{\mu}.$$

each component of $\boldsymbol{\mu}$ represents the average value of one coordinate of $\mathbf{x}$

Covariance Matrix of Random Vectors

From variance to covariance

variance measures the **spread** of a scalar random variable. For random vectors, we also care about how different components **vary together**

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Definition 3 (Covariance matrix)

for $\mathbf{x} \in \mathbb{R}^d$ with mean $\boldsymbol{\mu} = \mathbb{E}[\mathbf{x}]$,

$$\text{Cov}(\mathbf{x}) = \mathbb{E}[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\top] = \mathbb{E}[\mathbf{x}\mathbf{x}^\top] - \boldsymbol{\mu}\boldsymbol{\mu}^\top.$$

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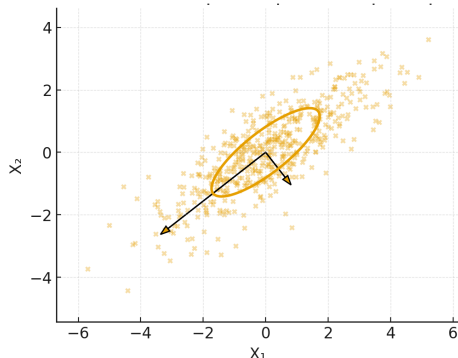
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$\text{Cov}(\mathbf{x})$ summarizes both the variability of each component (on the diagonal) and the **co-variability** between pairs of components (off the diagonal)

Covariance – geometric intuition



Geometric interpretation

The covariance matrix Σ defines the **shape** and **orientation** of the cloud of points of $\mathbf{x} = (X_1, X_2)$

Covariance Matrix – properties and entries

Main properties

- $\text{Cov}(\mathbf{x})$ is a $d \times d$ **symmetric, positive semidefinite** matrix.
- Its entries are given by

$$\text{Cov}(\mathbf{x})_{ij} = \mathbb{E}[(X_i - \mathbb{E}[X_i])(X_j - \mathbb{E}[X_j])].$$

- The related **second-moment matrix** is

$$\Sigma(\mathbf{x}) = \mathbb{E}[\mathbf{x}\mathbf{x}^\top], \quad \text{so that} \quad \text{Cov}(\mathbf{x}) = \Sigma(\mathbf{x}) - \boldsymbol{\mu}\boldsymbol{\mu}^\top.$$

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$\text{Cov}(\mathbf{x})$ contains all **second-order information** about $\mathbf{x}$: it tells us how mass is distributed and oriented around the mean.

Example: Multivariate Gaussian

Definition

a random vector $\mathbf{x} \in \mathbb{R}^d$ follows a **multivariate Gaussian** distribution if

$$\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}),$$

with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$.

If $\boldsymbol{\Sigma}$ is positive definite,

$$f_{\mathbf{x}}(x) = \frac{1}{\sqrt{(2\pi)^d |\boldsymbol{\Sigma}|}} \exp\left(-\frac{1}{2}(x - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(x - \boldsymbol{\mu})\right)$$

Example: Standard Normal Distribution

Definition

a random vector $\mathbf{g} = (g_1, \dots, g_d)$ is said to follow the **standard normal distribution**, written

$$\mathbf{g} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d),$$

if $g_1, \dots, g_d \sim \mathcal{N}(0, 1)$ are independent.

Probability density

$$f_{\mathbf{g}}(x) = \frac{1}{(2\pi)^{d/2}} \exp\left(-\frac{1}{2}\|x\|_2^2\right).$$

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The density depends only on $\|x\|_2$: it is perfectly **rotation invariant**.

Isotropy – definition

Definition 4 (isotropic random vectors)

a random vector $\mathbf{x} \in \mathbb{R}^d$ is called **isotropic** if its second-moment matrix is the identity matrix in $\mathbb{R}^d$, i.e. $\Sigma(\mathbf{x}) = \mathbf{I}_d$

an isotropic distribution has no preferred direction – it spreads equally in all directions, just like the standard normal

Isotropy – characterization lemma

Lemma 1 (characterization of isotropy)

a random vector $X \in \mathbb{R}^d$ is isotropic if and only if

$$\mathbb{E}\langle X, \mathbf{y} \rangle^2 = \|\mathbf{y}\|_2^2 \quad \forall \mathbf{y} \in \mathbb{R}^d$$

this means that all one-dimensional projections (directions) of X have the same expected energy – X has no preferred orientation

Random Vectors – equivalence block

$$\Sigma(X) = \mathbf{I}_d \iff \mathbb{E}\langle X, \mathbf{y} \rangle^2 = \|\mathbf{y}\|_2^2 \quad \forall \mathbf{y} \in \mathbb{R}^d$$

proof:

first note that two symmetric $d \times d$ matrices $\mathbf{A}$ and $\mathbf{B}$ are equal if and only if $\mathbf{y}^\top \mathbf{A} \mathbf{y} = \mathbf{y}^\top \mathbf{B} \mathbf{y}$ for all $\mathbf{y} \in \mathbb{R}^d$

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[hint: consider $\mathbf{y}^\top (\mathbf{A} - \mathbf{B}) \mathbf{y} = 0$ for certain choices of $\mathbf{y} \in \mathbb{R}^d$ and then use symmetry]

Isotropy – proof details continued

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notice that:

- $$\mathbb{E}\langle X, \mathbf{y} \rangle^2 = \mathbb{E}\langle \mathbf{y}, X \rangle \langle X, \mathbf{y} \rangle = \mathbb{E}[\mathbf{y}^\top X X^\top \mathbf{y}] = \mathbf{y}^\top (\mathbb{E}[X X^\top]) \mathbf{y}$$

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thus X is isotropic if and only if for all $\mathbf{y} \in \mathbb{R}^d$,

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$$\mathbf{y}^\top (\mathbb{E}[X X^\top]) \mathbf{y} = \mathbf{y}^\top \mathbf{I}_d \mathbf{y},$$

which is equivalent to $\Sigma(X) (= \mathbb{E}[X X^\top]) = \mathbf{I}_d$

□

Lemma 2

let $\mathbf{x}$ be an isotropic random vector in $\mathbb{R}^d$. Then

$$\mathbb{E}[\|\mathbf{x}\|_2^2] = d$$

moreover, if $\mathbf{x}$ and $\mathbf{y}$ are two independent isotropic random vectors in $\mathbb{R}^d$, then

$$\mathbb{E}\langle \mathbf{x}, \mathbf{y} \rangle^2 = d$$

Isotropy – proof (part i)

(i) $\mathbf{x}$ isotropic in $\mathbb{R}^d \Rightarrow \mathbb{E}[\|\mathbf{x}\|_2^2] = d$

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proof:

$$\begin{aligned}\mathbb{E} \|\mathbf{x}\|_2^2 &= \mathbb{E} \langle \mathbf{x}, \mathbf{x} \rangle = \mathbb{E}(\mathbf{x}^\top \mathbf{x}) = \mathbb{E} \operatorname{tr}(\mathbf{x}^\top \mathbf{x}) \\ &= \mathbb{E} \operatorname{tr}(\mathbf{x} \mathbf{x}^\top) \quad \text{cyclic property of the trace} \\ &= \operatorname{tr}(\mathbb{E}[\mathbf{x} \mathbf{x}^\top]) \quad \text{linearity trace} \\ &= \operatorname{tr}(\mathbf{I}_d) \quad \text{isotropy property} \\ &= d.\end{aligned}$$

□

Isotropy – proof (part ii)

(ii) $\mathbf{x}$ and $\mathbf{y}$ are two independent isotropic random vectors in $\mathbb{R}^d \Rightarrow \mathbb{E}\langle \mathbf{x}, \mathbf{y} \rangle^2 = d$

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Isotropy – proof (part ii)

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proof:

- denote the conditional expectation w.r.t. $\mathbf{x}$ as $\mathbb{E}_{\mathbf{x}}$

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proof:

- denote the conditional expectation w.r.t. $\mathbf{x}$ as $\mathbb{E}_{\mathbf{x}}$
- fixing the variable $\mathbf{y}$, and taking the conditional expectation w.r.t. $\mathbf{x}$, the **law of total expectation** states that

$$\begin{aligned}\mathbb{E}\langle \mathbf{x}, \mathbf{y} \rangle^2 &= \mathbb{E}_{\mathbf{y}} \mathbb{E}_{\mathbf{x}}[\langle \mathbf{x}, \mathbf{y} \rangle^2 \mid \mathbf{y}] \\ &\stackrel{(i)}{=} \mathbb{E}_{\mathbf{y}} \|\mathbf{y}\|_2^2 = d\end{aligned}$$

where (i) is due to Lemma 1

□